A sharp Weyl remainder for constant-index transmission eigenvalues

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Abstract

We prove a Weyl formula with remainder $O(r^{d-1})$ for the interior transmission problem with a constant refractive index $n > 1$ in an arbitrary bounded smooth domain in $\mathbb{R}^d$, $d \geq 2$. The count includes every nonzero complex energy of modulus at most r^2, with its algebraic multiplicity for the original maximal-domain transmission operator. The leading coefficient is $\omega_d(2\pi)^{-d}|\Omega|(1+n^{d/2})$. The argument combines the known Dirichlet-to-Neumann approximation with a sector estimate for a normalized boundary family. At inverse frequency scale h, its trace-class part has norm $O(h^{1-d})$ at scaled energy height h, and the sector estimate controls an actual logarithm of its determinant. Pole cancellation in a strip then gives the same order for the contour argument, including the endpoint contribution at every counting radius. No convexity assumption is used.

1 Introduction

The interior transmission problem asks when two Helmholtz fields with different wave speeds can have the same Cauchy data on the boundary. For a constant refractive index $n > 1$, it takes the form

$$\Delta u + n\lambda u = 0, \qquad \Delta v + \lambda v = 0 \quad \text{in } \Omega, \qquad u = v, \quad \partial_\nu u = \partial_\nu v \quad \text{on } \partial\Omega. \tag{1.1}$$

The coupling makes the spectral problem nonselfadjoint. Its high-energy count is nevertheless expected to be the sum of the two scalar Dirichlet counts, up to an error of boundary order. The main difficulty is to control the argument of a meromorphic boundary determinant close to the real spectrum, where the two Dirichlet spectra may meet the transmission spectrum.

Petkov and Vodev [4] relate the transmission count, with algebraic multiplicity, to the scalar Dirichlet counts through an exact boundary index formula. Their construction reduces the counting remainder to estimates for a meromorphic Dirichlet-to-Neumann family. Vodev's high-frequency approximation [8] supplies, in the isotropic case, a fixed eigenvalue-free strip in the wave-number plane and a counting remainder $O_\epsilon(r^{d-1+\epsilon})$ for every $\epsilon > 0$. The boundary-order remainder $O(r^{d-1})$ is the conjectured endpoint [7, Section 7, Conjecture 1, (19)]. We prove that endpoint for the constant-index problem (1.1) on every smooth bounded domain.

We first specify the realization and the count. All spaces are complex. Let $\Omega \subset \mathbb{R}^d$ be bounded and connected, with C^∞ boundary. On $\mathcal{H} = L^2(\Omega) \oplus L^2(\Omega)$, let

$$A(u,v) = (-n^{-1}\Delta u, -\Delta v), \qquad \begin{aligned} \mathcal{D}(A) = \{(u,v) \in \mathcal{H} : \Delta u, \Delta v \in L^2(\Omega), \\ \gamma_0 u = \gamma_0 v, \quad \gamma_1 u = \gamma_1 v\}. \end{aligned} \tag{1.2}$$

Here the Laplacians are distributional and γ_0, γ_1 are the generalized Dirichlet and normal traces on the maximal Laplace graph domain. Either common choice of normal gives this

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same domain. This is the closed transmission realization used throughout the paper. Its nonzero eigenvalues are isolated and have finite algebraic multiplicity. For a small positively oriented circle C_λ enclosing one such eigenvalue, set

$$m_T(\lambda) = \operatorname{rank} \frac{1}{2\pi i} \int_{C_\lambda} (z - A)^{-1}\, dz, \qquad N(r) = \sum_{0<|\lambda|\le r^2} m_T(\lambda). \tag{1.3}$$

Thus $N(r)$ counts energies, including nonreal energies, with the Riesz multiplicity of the original operator. It does not count the two square roots of an energy separately. Finiteness near zero is established in lemma 2.1.

Theorem 1.1. *Let $d \ge 2$, let $\Omega \subset \mathbb{R}^d$ be a bounded connected domain with C^∞ boundary, and fix a constant $n > 1$. For the operator (1.2) and count (1.3),*

$$N(r) = \frac{\omega_d}{(2\pi)^d} |\Omega| (1 + n^{d/2}) r^d + O_{\Omega,n}(r^{d-1}) \qquad (r \longrightarrow \infty). \tag{1.4}$$

Here ω_d is the volume of the unit ball in $\mathbb{R}^d$ and $|\Omega|$ is the volume of Ω. The estimate holds for every sufficiently large real radius r, including radii through eigenvalues.

The boundary may have several components; no convexity or other geometric restriction is imposed. The index n is fixed, and the constant in the remainder may deteriorate as $n \downarrow 1$. The theorem is the constant-index isotropic case of the conjecture cited above. It makes no assertion for variable refractive indices or anisotropic principal parts.

The proof retains the Poisson construction and the exact Riesz/index identity of [4], and the approximation of [8]. The additional estimate concerns the determinant's phase at the critical height $|\Im\zeta| \asymp h$ in the scaled energy $\zeta = h^2\lambda$, where $h \asymp r^{-1}$ is the inverse counting scale. After multiplication by one tangential derivative and a scalar factor, the difference of the two Dirichlet-to-Neumann maps has principal symbol

$$\frac{2\sqrt{1 + r_0}}{\sqrt{r_0 - n\zeta} + \sqrt{r_0 - \zeta}},$$

with square roots of positive real part in the upper half-plane. Here r_0 is the symbol of the positive boundary Laplacian. This symbol stays in a fixed sector and tends to one at high tangential frequency. The full approximation in its one-derivative output norm therefore gives a uniformly coercive boundary family at height Ch. A holomorphic high-frequency normalization leaves a trace-class perturbation of the identity of size $O(h^{1-d})$. Coercivity along the straight-line homotopy to the identity bounds an analytic logarithm of its determinant, including its imaginary part. This is the phase estimate needed at the endpoint.

The determinant still has Dirichlet poles inside the thin strip. We cancel their divisors with finite strip factors and use a harmonic barrier to transfer the horizontal bound into the strip. Jensen's formula bounds the local transmission multiplicity by $O(h^{1-d})$; a disk factorization then bounds the argument on each short vertical side by the same order. The index formula gives the annular count, and the local estimate removes contour shifts at exact endpoints. Dyadic summation yields (1.4).

Section 2 records the realization, scaling and external boundary estimates. Section 3 constructs the normalized determinant and its logarithm, and section 4 proves the thin-strip estimates. Section 5 completes the count. The radial calculation in section A, related to the constant-index models in [5], illustrates why common Dirichlet values must retain their original algebraic multiplicity.

2 Spectral and boundary preliminaries

Throughout the proof, d, Ω and n satisfy the hypotheses of theorem 1.1, and $q = d - 1$. Constants may depend on these fixed data and on fixed cutoffs and compact energy sets. They are independent of the small parameter h and of the spectral point in the region stated. No uniformity as $n \downarrow 1$ is asserted.

2.1 The realization and the two spectral parameters

The realization of the transmission operator used by Petkov and Vodev [4, Section 3] acts on $L^2(\Omega, n\,dx) \oplus L^2(\Omega, dx)$ with the same maximal graph domain as A. The identity map to $L^2(\Omega)^2$ is a bounded isomorphism, since n is constant. It conjugates the operators and their resolvents, so it preserves Riesz projection ranks. We shall use their counting identity with precisely this multiplicity.

We write λ for an energy eigenvalue and use

$$\zeta = h^2\lambda, \qquad k = \sqrt{\lambda}, \qquad \Re k \geq 0. \tag{2.1}$$

The wave number k is used only to locate the spectrum. The counting function always counts energies, once each.

Lemma 2.1. *Every eigenfunction of A at a nonzero eigenvalue is smooth up to the boundary. There is a punctured neighborhood of zero containing no nonzero eigenvalues. In particular, the nonzero counting function is finite on bounded intervals of radii.*

Proof. Let (u, v) be an eigenfunction at $\lambda \neq 0$, and set $w = u - v$. Then $w, \Delta w \in L^2(\Omega)$ and both generalized Cauchy traces of w vanish. Zero-Dirichlet elliptic regularity gives $w \in H^2(\Omega)$; the two zero traces then give $w \in H_0^2(\Omega)$. To apply the regularity statement at this initial graph regularity, subtract the zero-Dirichlet solution with the same $(1 - \Delta)w$. The remaining homogeneous solution has zero generalized Dirichlet trace and vanishes by Green's identity against smooth Dirichlet inverse test functions. The usual regularity theorem therefore applies. We use the trace and elliptic regularity statements in [3, Chapter 11, especially Theorem 11.14].

The equations imply

$$u = -\frac{\Delta w + \lambda w}{(n-1)\lambda}, \qquad v = -\frac{\Delta w + n\lambda w}{(n-1)\lambda}, \tag{2.2}$$

and hence

$$\Delta^2 w + (n+1)\lambda\Delta w + n\lambda^2 w = 0, \qquad w \in H_0^2(\Omega). \tag{2.3}$$

Here $w \neq 0$, since $u = v$ would force $(n-1)\lambda u = 0$. The right-hand side of the fourth-order equation is initially in L^2. Clamped elliptic regularity gives $w \in H^4$ and then, by iteration, $w \in C^\infty(\overline{\Omega})$. Formula (2.2) gives the same conclusion for u, v.

On $H_0^2(\Omega)$, the norm $\|\Delta w\|_2$ is equivalent to the H^2 norm. Indeed, integration by parts for compactly supported smooth approximants gives $\sum_{i,j} \|\partial_i\partial_j w\|_2^2 = \|\Delta w\|_2^2$, and Poincare's inequality controls the lower derivatives. Testing (2.3) against w therefore shows that it has no nonzero solution for sufficiently small $|\lambda|$. Equivalently, its weak form is an invertible form at $\lambda = 0$ plus $\lambda C_1 + \lambda^2 C_2$, with C_1, C_2 bounded in that Hilbert norm. Away from zero we use the qualitative discreteness and finite-rank meromorphic resolvent construction of [4, Theorem 1.1 and Section 3]. This proves the bounded-radius assertion. No compact-resolvent assertion for A at zero is needed. □

Vodev's eigenvalue-free region applies to the smooth eigenfunctions just obtained. In the isotropic case, [8, Theorem 2.1 and the following remark] gives a constant C_k such that all sufficiently large transmission wave numbers satisfy

$$|\Im k| \le C_k. \tag{2.4}$$

The remark is used for $0 \le \Re k \le 1$; Theorem 2.1 supplies the region $\Re k > 1$. Thus (2.4) covers all large complex energies, including the exclusion of a large negative-energy branch.

For $m = 1, n$, let $G_{m,D} = -m^{-1}\Delta_D$ be the positive Dirichlet operator, and let $D_m(t)$ count its eigenvalues at most t. The sharp scalar Dirichlet Weyl estimate [6, (0.1)] gives

$$D_m(t) = \tau_m t^{d/2} + O(t^{q/2}), \qquad \tau_m = \frac{\omega_d}{(2\pi)^d}|\Omega| m^{d/2}. \tag{2.5}$$

The scaling follows from $D_m(t) = D_1(mt)$. Consequently, in a fixed compact positive scaled-energy interval, every interval of length Ch has combined scaled Dirichlet multiplicity $O(h^{-q})$, for fixed C. To see this, subtract (2.5) at the endpoints divided by h^2. The leading difference and both remainders are $O(h^{-q})$. Enlarging the interval by h proves the same assertion for closed endpoints and for a single eigenvalue's multiplicity.

2.2 Dirichlet-to-Neumann maps and the source approximation

Let $\Gamma = \partial\Omega$, and use the inner unit normal on Γ. For ζ outside the two scaled Dirichlet spectra, define

$$\mathcal{N}_m(\zeta, h) f = h\partial_{\nu_{\text{in}}} u|_\Gamma, \qquad (h^2\Delta + m\zeta)u = 0, \quad u|_\Gamma = f. \tag{2.6}$$

The families extend meromorphically, with simple finite-rank poles at scaled Dirichlet values. Put

$$\Lambda_h = (I - h^2\Delta_\Gamma)^{1/2}, \qquad \|f\|_{H^1_h(\Gamma)} = \|\Lambda_h f\|_{L^2(\Gamma)}. \tag{2.7}$$

This norm is uniformly equivalent to the local semiclassical Sobolev norm. The boundary Laplacian is nonpositive; the square root is its positive spectral one. Write $r_0(x', \xi')$ for the principal symbol of $-\Delta_\Gamma$, and $\ell = (1 + r_0)^{1/2}$.

We use semiclassical quantization with local phase $e^{-i\langle x'-y', \xi'\rangle/h}$ and factor $(2\pi h)^{-q}$, as in [8]. The opposite phase is obtained by $\xi' \mapsto -\xi'$. It gives the same symbol estimates and the same principal symbols here, since r_0 is even in ξ'. We use a linear partition-of-unity quantization with $\mathrm{Op}_h(1) = I$.

The approximation from [8, Theorem 1.2] will be used in its full mapping norm. Work first with a semiclassical parameter h_e such that $\Re((h_e k)^2) = 1$. For every fixed sufficiently small $\delta > 0$, let $\chi_{m,\delta}$ be the source cutoff equal to one in a δ^2 neighborhood of the glancing set $\{r_0 = m\}$ and supported in a slightly larger such neighborhood. With

$$\rho_m = \sqrt{r_0 - m(h_e k)^2}, \qquad \Re\rho_m < 0,$$

the source gives

$$\left\| h_e N(k; m) - \mathrm{Op}_{h_e}\left(\rho_m(1 - \chi_{m,\delta}) + h_e b\right) \right\|_{L^2 \to H^1_{h_e}} \le C\delta. \tag{2.8}$$

Here $N(k; m)$ is the unscaled inner-normal map for $(\Delta + mk^2)u = 0$. Here S^0_0 denotes symbols of order zero with standard symbol seminorms uniform in the semiclassical parameter. The symbol $b \in S^0_0$ is independent of both k and the index function. For each $0 < \epsilon < 1$, the estimate holds when

$$C_\delta \le |\Im k| \le (\Re k)^{1-\epsilon}, \qquad \Re k \ge C_{\epsilon,\delta}.$$

The domain may be any smooth bounded domain. No concavity is part of this theorem. The index independence of b and the H^1 output in (2.8) are both needed below.

We also use the closed-manifold symbol calculus, boundedness and Garding estimates, in the conventions of [2, Chapter 4 and Appendix E]. For fixed cutoffs, their symbol seminorms are uniform in h. The positive operator in (2.7) has symbol $\ell + hS^0$. One way to justify this for the unbounded square root is to start from the norm-convergent integral

$$(I - h^2\Delta_\Gamma)^{-1/2} = \frac{2}{\pi}\int_0^\infty (I - h^2\Delta_\Gamma + t^2)^{-1}\, dt. \tag{2.9}$$

The elliptic inverse starts with $(1 + r_0 + t^2)^{-1}$. Its coefficient at order h^j has joint order $-2-j$ in (ξ', t), by the usual composition recursion. Integrating in t gives order $-1-j$, with principal symbol ℓ^{-1}. An arbitrarily long recursion makes the remainder integrable in any fixed Sobolev seminorm; the exact positive resolvent bounds the correction. Multiplication by $I - h^2\Delta_\Gamma$ gives the stated symbol of Λ_h. In particular, $\Lambda_h : H^1_h \to L^2$ has norm one, and $\Lambda_h : H^{s+1} \to H^s$ is bounded uniformly for $h \le 1$ at each fixed ordinary Sobolev order.

3 A normalized boundary determinant

The normalization is chosen so that its principal symbol lies in a fixed sector and tends to one at high tangential frequency. The first property will control the logarithm; the second will make its trace-class part have boundary-order size. Define

$$T_h = \mathcal{N}_n - \mathcal{N}_1, \qquad B_h(\zeta) = \frac{2}{(n-1)\zeta}\Lambda_h T_h(\zeta). \tag{3.1}$$

For each fixed h, this is a bounded meromorphic family on boundary L^2. Its high-frequency order is zero, since the first-order terms of the two Dirichlet-to-Neumann maps cancel. Fix a compact rectangle $\mathcal{D}$ inside $\{1/2 < \Re\zeta < 3,\ |\Im\zeta| < 9/10\}$, whose interior contains $[3/5, 14/5]$. Every later contour lies in its interior for small h.

3.1 A uniform sector at height h

Lemma 3.1. *There are constants $C_*, c_B, C_B > 0$ such that, for every fixed $H > C_*$ and sufficiently small h, the following estimates hold uniformly for $3/5 \le x = \Re\zeta \le 14/5$ and $C_*h \le |y| = |\Im\zeta| \le Hh$:*

$$\|B_h(\zeta)\| \le C_B, \qquad \Re(e^{-i\theta}\langle B_h(\zeta)v, v\rangle) \ge c_B\|v\|_2^2, \quad \theta = \operatorname{sgn}(y)\pi/4. \tag{3.2}$$

The constants C_*, c_B, C_B can be fixed before H; the smallness threshold for h may then depend on H.

Proof. Suppose first that $y > 0$, and choose

$$\alpha_m = \sqrt{r_0 - m\zeta}, \qquad \Re\alpha_m > 0, \quad \Im\alpha_m < 0.$$

The inner-normal principal symbol is $-\alpha_m$. Thus the principal symbol of (3.1) is

$$a = \frac{2\ell(\alpha_1 - \alpha_n)}{(n-1)\zeta} = \frac{2\ell}{\alpha_n + \alpha_1}. \tag{3.3}$$

The two roots lie in the fourth quadrant, so a lies in the first. Moreover,

$$0 < c_0 \le |a| \le C_0, \qquad \Re(e^{-i\pi/4}a) \ge c_0/\sqrt{2}. \tag{3.4}$$

For large r_0, the denominator is comparable to $2\sqrt{r_0}$. On a bounded frequency set, the identity $\alpha_1^2 - \alpha_n^2 = (n-1)\zeta$ bounds at least one root away from zero. Vectors in the same quadrant cannot cancel, proving the remaining bounds in (3.4).

For x in the stated compact interval, put $h_e = h/\sqrt{x}$ in (2.8). Then $\Re((h_e k)^2) = 1$. The change $\xi' = \sqrt{x}\eta$ gives $\mathrm{Op}_{h_e}(c(\eta)) = \mathrm{Op}_h(c(\xi'/\sqrt{x}))$. Multiplication of (2.8) by $\sqrt{x}$ transforms its root to $-\alpha_m$ and its common correction to $hb(x', \xi'/\sqrt{x})$. This correction cancels between the two indices. The two H^1 norms are uniformly equivalent. Also

$$\Im k = \frac{y}{2h\Re\sqrt{\zeta}}.$$

Choosing C_* sufficiently large meets the lower source bound; for fixed H the upper source bound holds for small h, for example with $\epsilon = 1/2$.

On a deleted neighborhood $|r_0 - mx| \le C\delta^2$, one has $|\alpha_m| \le C(\delta + \sqrt{Hh})$. For fixed δ and small h, deleting it therefore changes the symbol in (3.3) by $O(\delta)$ in supremum norm. The deleted frequencies are bounded. Outside them the cut symbols have uniform h seminorms, with constants depending on δ. Their difference has order -1 at infinity. Compose with Λ_h and use the actual $L^2 \to H^1_h$ error in (2.8). It becomes an $O(\delta)$ error on L^2. Composition and Garding for the smooth cut symbol give

$$\Re(e^{-i\pi/4}\langle B_h v, v\rangle) \ge (c_0/\sqrt{2} - C\delta - C_\delta h)\|v\|_2^2,$$

as well as the upper norm bound. Fix δ first and then h. Conjugating the real-coefficient boundary problems gives the lower-half estimate with angle $-\pi/4$. The auxiliary cut symbols need not be holomorphic: the estimates concern the exact family B_h. □

3.2 Holomorphic normalization and trace norm

Let $\chi_h = \mathrm{Op}_h(\chi)$, where χ equals one for $r_0 \le R$ and zero for $r_0 \ge 2R$. The fixed R is chosen above all glancing frequencies on $\mathcal{D}$. The elliptic Poisson construction in [4, Section 2, (2.7), (2.9), (2.13)], at an arbitrarily large fixed order J, gives a holomorphic approximation Q_h to $B_h(I - \chi_h)$. We take the actual composition with Λ_h, without discarding any further composition remainder, and set

$$P_h = Q_h + \chi_h. \tag{3.5}$$

The local Poisson coordinate in that construction is $x_1 > 0$ inward. With $D_{x_1} = -ih\partial_{x_1}$, our inner derivative is $h\partial_{x_1} = iD_{x_1}$. The source phase has $\rho = i\sqrt{r_0 - m\zeta}$, and hence its actual inner-normal symbol is $i\rho = -\sqrt{r_0 - m\zeta}$. This convention fixes the sign in (3.3); the common change of normal in the transmission index identity has no effect.

The leading symbol of P_h is $a(1-\chi) + \chi$. At high frequency $a = 1 + O(r_0^{-1})$, with the same decay under the finite derivatives needed for boundedness. The first subprincipal difference of the two Poisson boundary symbols has order -2 when both leading differential coefficients are one; see the statement in the proof of [4, Lemma 3.3, p. 18, using (2.8)]. The coefficient at h^j, $j \ge 2$, has order at most $1 - j$. After Λ_h all these terms are bounded and carry a positive power of h. It follows that

$$\|P_h - I\| \le \varepsilon_R + C_R h, \qquad \varepsilon_R \longrightarrow 0 \quad (R \longrightarrow \infty), \tag{3.6}$$

uniformly on $\mathcal{D}$. Choose R and then h so that this norm is less than $\varepsilon_0 < 1/2$, where $2C_B\varepsilon_0 < c_B/2$. The Neumann series makes P_h^{-1} holomorphic on the full rectangle and uniformly bounded by two.

We next keep the smoothing error separate from the low frequencies. Let

$$\delta_D(\zeta, h) = \min\{1, \mathrm{dist}(\zeta, h^2 \operatorname{spec} G_{1,D} \cup h^2 \operatorname{spec} G_{n,D})\}.$$

The L^2 spectral theorem and ordinary zero-Dirichlet regularity give, for every fixed integer $j \geq 1$,

$$\|(h^2 G_{m,D} - \zeta)^{-1} f\|_{H^{2j}} \leq C_j h^{-2j} \delta_D^{-1} \|f\|_{H^{2j-2}}. \tag{3.7}$$

Indeed the L^2 norm is at most δ_D^{-1}, and iteration of

$$G_{m,D}(h^2 G_{m,D} - \zeta)^{-1} f = h^{-2}(f + \zeta(h^2 G_{m,D} - \zeta)^{-1} f)$$

costs h^{-2} at each step of the elliptic estimate. It does not add another distance factor. In particular, this argument does not require a higher ordinary-Sobolev resolvent bound uniform in h.[1]

The exact corrected Poisson solution gives

$$B_h - P_h = (B_h - I)\chi_h + E_{J,h}. \tag{3.8}$$

Fix an integer $s > q/2$. The pure interior error in [4, (2.13)], with its index $s+1$, maps boundary L^2 to $H^{2s+7/2}(\Omega)$ with norm $O(h^{J-2s-2-\ell_1})$, where ℓ_1 is independent of s, J and $s+1 \leq J/4$. Embed into H^{2s+2} and use (3.7) with $j = s+2$. The result is in $H^{2s+4}(\Omega)$, at a cost $h^{-2s-4}\delta_D^{-1}$. The scaled normal trace has output $H^{2s+5/2}(\Gamma)$ and adds a factor h. Finally Λ_h maps to $H^{2s+3/2}(\Gamma)$, which embeds into $H^{2s}(\Gamma)$. Thus

$$\|E_{J,h}\|_{L^2 \to H^{2s}} \leq C_{s,J} h^{J-4s-5-\ell_1} \delta_D^{-1}. \tag{3.9}$$

Choose $J \geq 4(s+1)$ and $J > 4s+6+\ell_1$. At height $|y| \geq C_* h$, the right-hand side is bounded. The embedding $H^{2s}(\Gamma) \to L^2$ is trace class, since $2s > q$.

The low-frequency cutoff has

$$\|\chi_h\|_1 = O(h^{-q}). \tag{3.10}$$

For completeness, its norm on L^2 is $O(1)$ and its norm into H^{2s} is $O(h^{-2s})$. The singular values of the fixed embedding are $O(j^{-2s/q})$; see [4, (2.11)] or [2, Appendix E]. Hence $s_j(\chi_h) \leq C_s \min(1, h^{-2s} j^{-2s/q})$. Summing below and above $j \asymp h^{-q}$ proves (3.10).

Combining lemma 3.1 with (3.8)– (3.10), define

$$F_h = B_h P_h^{-1} = I + K_h, \qquad g_h = \det(I + K_h), \qquad M = h^{-q}. \tag{3.11}$$

On the critical strips $C_* h \leq |y| \leq Hh$,

$$\|K_h\|_1 \leq CM, \qquad \Re(e^{-i\theta}\langle F_h v, v\rangle) \geq (c_B/2)\|v\|_2^2. \tag{3.12}$$

For the second estimate use $\|F_h - B_h\| \leq 2C_B\varepsilon_0 < c_B/2$. The family is meromorphic in trace norm throughout $\mathcal{D}$, with finite-rank principal parts. We shall also use a coarse bound

$$\log|g_h(\zeta)| \leq \|K_h(\zeta)\|_1 \leq C h^{-A_0} \delta_D^{-1} \tag{3.13}$$

for one fixed finite A_0. To obtain it, apply a fixed smooth boundary extension to $\chi_h f$, subtract the Dirichlet resolvent of its equation residual, and use (3.7) at a fixed sufficiently large order. Cutoff differentiation and each fixed trace cost only finitely many powers of h^{-1}. The high-frequency remainder is already covered by (3.9). Factoring through the trace-class Sobolev embedding gives (3.13). Only this polynomial form will be needed at remote vertical edges.

[1] We do not use (2.3) of [4]. Its displayed h factor fails for the single Dirichlet-to-Neumann map on the unit three-ball at scaled energy -1 with boundary datum 1: the solution is $\sinh(r/h)/(r\sinh(1/h))$, and the convention $-ih\partial_{\nu_{\text{out}}}$ gives $-i(\coth(1/h) - h)$. The truncated distance to the positive Dirichlet spectrum is one, but this value does not tend to zero. Only the pure Poisson construction, before that auxiliary estimate, is used here.

3.3 A complex logarithm and the divisor identity

The trace norm in (3.12) controls a particular analytic logarithm, including its imaginary part.

Lemma 3.2. *On each of the upper and lower critical strips there is a holomorphic function $L_h^{\pm}$ such that*

$$e^{L_h^{\pm}} = g_h, \qquad |L_h^{\pm}| \le CM, \qquad (L_h^{\pm})' = g_h'/g_h. \tag{3.14}$$

Proof. For $0 \le t \le 1$, $I + tK_h = (1-t)I + tF_h$. Its rotated real part is bounded below by $\min(\cos\theta, c_B/2)\|v\|_2^2$. This gives injectivity and closed range; the corresponding conjugate rotation for the adjoint gives dense range. Thus its inverse is uniformly bounded. Set

$$L_h^{\pm}(\zeta) = \int_0^1 \operatorname{Tr}\left((I + tK_h)^{-1} K_h\right) dt.$$

The integrand is holomorphic and uniformly bounded in trace norm by CM. The determinant differentiation formula, integrated from $t = 0$ where the determinant is one, proves (3.14). These trace-class formulas can also be obtained directly by finite-rank approximation: the exterior-power determinant series is absolutely bounded by $e^{\|K_h\|_1}$, converges locally uniformly in trace norm, and has the finite-dimensional differentiation formula where invertible. Cauchy's formula passes its derivatives to the limit. □

For a scaled Dirichlet value p, let $m_D(p)$ be the sum of its two Dirichlet multiplicities. Let $m_T(p)$ be the Riesz multiplicity of p/h^2 for A, or zero if it is not a transmission eigenvalue. Away from the Dirichlet spectra put $m_D = 0$ as well. Orders of zeros are positive and orders of poles negative.

Proposition 3.3. *The determinant in (3.11) satisfies*

$$\operatorname{ord}_p g_h = m_T(p) - m_D(p). \tag{3.15}$$

Equivalently, for every simple positively oriented contour γ in $\mathcal{D}$ avoiding the transmission and Dirichlet spectra,

$$\#_T(\gamma) = \#_{D,1}(\gamma) + \#_{D,n}(\gamma) + \frac{1}{2\pi i}\int_\gamma \frac{g_h'}{g_h}\, d\zeta. \tag{3.16}$$

All transmission counts here use the original Riesz ranks, including at common Dirichlet values.

Proof. The original-operator trace identity is [4, Proposition 3.1, Corollary 3.2 and Lemma 3.6]. Its meromorphic Fredholm hypothesis holds here by (3.8) and the invertible critical-strip values in (3.12). The identity between the constant-weighted source Hilbert space and our unweighted L^2 space is a bounded conjugacy, so it preserves the resolvents and Riesz ranks; see section 2.

It remains to check the normalizations. Passing to B_h multiplies the boundary difference on the left by a holomorphic nonzero scalar, a fixed common normal sign and the invertible map $\Lambda_h : H_h^1 \to L^2$. Passing to F_h multiplies it on the right by the holomorphic invertible P_h^{-1}. None changes the local meromorphic index. Indeed, finite-dimensional projections separate an invertible complementary block from the finite singular block of a Fredholm family. Holomorphic invertible changes multiply its Schur complement by holomorphic invertible factors, whose determinant orders are zero. For an identity-plus-trace-class representative the remaining order is that of its Fredholm determinant. The source identity therefore gives (3.16), and small circles give (3.15). In particular, pole order is at most $m_D(p)$, even when a transmission point coincides with one or both Dirichlet spectra. □

4 Pole cancellation in a thin strip

We now use $g_h = \det(I + K_h)$ from (3.11) and $M = h^{1-d}$. Its only possible poles are real Dirichlet values; its upper and lower logarithms are bounded by CM at imaginary height comparable to h. The purpose of the pole factors is to use those two bounds across the real axis.

Fix

$$L > \max\{2C_*, 8(C_k + 1)\}, \qquad H > 20L. \tag{4.1}$$

These constants are chosen before reducing h. Recall from (2.4) that $|\Im\sqrt{\lambda}| \le C_k$ and $\zeta = h^2\lambda$. In the band $\Re\zeta \le 3$ this gives

$$|\Im\zeta| \le 2C_k\sqrt{3 + C_k^2h^2}\, h < (L/2)h$$

for all sufficiently large eigenvalues and small h. Choose $a_h \in [3/5, 7/10]$ and $b_h \in [27/10, 14/5]$ at distance at least h^{d+2} from the scaled Dirichlet spectra. Such choices exist: there are $O(h^{-d})$ relevant Dirichlet values counted with multiplicity, so the excluded intervals have total length $O(h^2)$. Set

$$S_h = \{a_h < \Re\zeta < b_h,\ |\Im\zeta| < Hh\}, \qquad b_p(\zeta) = \tanh\frac{\pi(\zeta - p)}{4Hh},$$

and form the finite product

$$\mathcal{B}_h(\zeta) = \prod_{a_h<p<b_h} b_p(\zeta)^{m_D(p)}, \qquad f_h = g_h\mathcal{B}_h. \tag{4.2}$$

Each factor is analytic in the strip, has just the simple zero p there, and satisfies $|b_p| \le 1$ inside and $|b_p| = 1$ on its horizontal edges. These assertions follow from

$$|\tanh(u + iv)|^2 = \frac{\sinh^2 u + \sin^2 v}{\sinh^2 u + \cos^2 v}, \qquad |v| \le \pi/4. \tag{4.3}$$

The nearest poles are at $p \pm 2iHh$. By (3.15), f_h is holomorphic on a neighborhood of the closed rectangle. Its zero orders at all transmission points, including common Dirichlet points, are exactly m_T.

4.1 Upper bounds and local counts

On the horizontal edges lemma 3.2 gives $\log|f_h| \le CM$. On the vertical edges (3.13) and the chosen distance give the coarser bound Ch^{-A_0-d-2}. This large bound has exponentially small influence in the interior. For a direct comparison, put $\kappa = \pi/(4Hh)$ and

$$v_h(x + iy) = \sqrt{2}\big(e^{-\kappa(x-a_h)} + e^{-\kappa(b_h-x)}\big)\cos(\kappa y).$$

It is harmonic and nonnegative on S_h, and is at least one on both vertical edges. Therefore $CM + Ch^{-A_0-d-2}v_h$ majorizes $\log|f_h|$. To justify this even at zeros, take a holomorphic function whose real part is the harmonic majorant and apply the maximum-modulus principle to f_h times its negative exponential. If $4/5 \le x \le 11/5$, then $v_h \le Ce^{-c/h}$. Hence

$$\log|f_h(\zeta)| \le CM, \qquad 4/5 \le \Re\zeta \le 11/5, \quad |\Im\zeta| \le Hh. \tag{4.4}$$

For a in a fixed compact inner interval containing $[1, 2]$, let $z_a = a + iLh$. The lower bound in (3.14) gives $\log|g_h(z_a)| \ge -CM$. Formula (4.3) also gives

$$0 \le -\log|b_p(z_a)| \le Ce^{-c|a-p|/h}. \tag{4.5}$$

For bounded $|a-p|/h$, the fixed positive imaginary argument $\pi L/(4H)$ bounds the modulus away from zero. For larger separation, use the exponential convergence of tanh to 1 or -1. Each interval of length h has combined scaled Dirichlet multiplicity $O(M)$, by (2.5). Partitioning the real axis into these intervals and summing the exponential weights yields

$$-\log|\mathcal{B}_h(z_a)| \le CM, \qquad \log|f_h(z_a)| \ge -CM. \tag{4.6}$$

The constants are uniform in a.

For small h, the disk of radius $8Lh$ about z_a lies in the region of (4.4). Jensen's formula and (4.6) bound the number of its zeros in radius $7Lh$ by CM: each such zero contributes at least $\log(8/7)$ to the Jensen sum. If the outer circle contains zeros, use nearby radii tending to $8Lh$. The smaller disk $|\zeta-a| \le 2Lh$ lies inside that disk. Since the zeros of f_h have precisely the transmission orders, we obtain

$$\sum_{|\zeta-a|\le 2Lh} m_T(\zeta) \le CM. \tag{4.7}$$

In particular, this bound holds at every real endpoint, without a spacing assumption. The elementary disk formulas used here and below follow from Cauchy's formula, the maximum principle and Schwarz's lemma; see [1, Sections 6–8, 13 and 17].

4.2 The argument on a short vertical segment

Lemma 4.1. *For every such a, there is an a' with $|a'-a| < h/10$ such that the segment $\gamma_{a'} = [a'-iLh, a'+iLh]$ avoids both spectra and*

$$\left|\Im \int_{\gamma_{a'}} \frac{g_h'}{g_h}\, d\zeta\right| \le CM. \tag{4.8}$$

Proof. Choose $R_a \in [6Lh, 7Lh]$ so that its circle about z_a contains no zero of f_h. Factor the zeros in this disk by its disk Blaschke product $\mathcal{Z}_h$, and write

$$f_h = \mathcal{Z}_h e^{w_h}.$$

The quotient is zero-free on a neighborhood of the closed disk, so its holomorphic logarithm w_h exists there. There are $O(M)$ factors. They have unit modulus on the circle and modulus at most one inside. Thus (4.4) and (4.6) imply

$$\Re w_h \le CM, \qquad \Re w_h(z_a) \ge -CM.$$

The usual real-part form of the Cauchy estimate gives

$$|w_h'(\zeta)| \le CM/h \quad \text{for } |\zeta - z_a| \le 4Lh. \tag{4.9}$$

For clarity, it follows by setting $W = U - w_h + i\Im w_h(z_a)$ with $U \ge \sup \Re w_h$. If $W(z_a) > 0$, the function $(W - W(z_a))/(W + W(z_a))$ maps the disk to the unit disk and vanishes at its center. Schwarz's lemma bounds W on a smaller disk by a constant times $W(z_a)$, and Cauchy's estimate bounds its derivative. If $W(z_a) = 0$, the real-part minimum principle makes it constant. The fixed radius ratios give (4.9).

Choose $|a'-a| < h/10$ avoiding the finitely many real parts of local zeros and all real Dirichlet values. The resulting segment is inside $|\zeta - z_a| < 4Lh$. For a linear zero factor $\zeta - w$, the change of argument is the angle subtended by a straight segment, so

$$\left|\Im \int_{\gamma_{a'}} \frac{d\zeta}{\zeta - w}\right| \le \pi.$$

In the coordinate $z = \zeta - z_a$, a disk factor has denominator $R_a^2 - \overline{w - z_a}\, z$; on the inner disk its logarithmic derivative is $O(h^{-1})$, uniformly in the zero. Its integral is therefore $O(1)$. Summing the $O(M)$ factors and (4.9) proves

$$\left| \Im \int_{\gamma_{a'}} f_h'/f_h \, d\zeta \right| \le CM.$$

It remains to subtract $\mathcal{B}_h'/\mathcal{B}_h$. For $|p - a| \le 10Hh$, the difference $b_p'/b_p - (\zeta - p)^{-1}$ is $O(h^{-1})$ on the segment, uniformly even as $a' - p$ tends to zero. This follows by scaling by h and removing its simple pole; the other zeros and poles lie outside the strip. There are $O(M)$ near factors. Their removed linear terms contribute at most π each. For the remaining factors,

$$|b_p'/b_p| \le Ch^{-1} e^{-c|a-p|/h}.$$

The length-h Dirichlet bound and summation of this exponential tail, followed by integration over length $2Lh$, give $O(M)$ again. Using (4.2) proves (4.8). □

5 The Weyl formula

We can now compare the transmission and scalar Dirichlet counts with boundary-order error. Apply lemma 4.1 at 1 and 2 to choose $a' = 1 + O(h)$ and $b' = 2 + O(h)$, with shifts less than $h/10$. Let $\mathcal{R}_h$ be the positively oriented rectangle with these vertical sides and horizontal sides at $\pm Lh$. Its boundary avoids the two spectra, including common Dirichlet/transmission points. On a horizontal side,

$$\int g_h'/g_h \, d\zeta = L_h^{\pm}(\text{end}) - L_h^{\pm}(\text{start}) = O(M)$$

by the actual analytic branch in lemma 3.2. The two short vertical sides satisfy (4.8). Since the complete contour integral divided by $2\pi i$ is real, these imaginary-part estimates and proposition 3.3 give

$$\#_T(\mathcal{R}_h) - \#_{D,1}(\mathcal{R}_h) - \#_{D,n}(\mathcal{R}_h) = O(h^{1-d}). \tag{5.1}$$

For every sufficiently large transmission energy in a fixed positive scaled band, (2.4) implies

$$|\Im\zeta| = O(h), \qquad |\zeta| - \Re\zeta = \frac{(\Im\zeta)^2}{|\zeta| + \Re\zeta} = O(h^2). \tag{5.2}$$

The choice (4.1) places all of them below $Lh/2$. Replacing $\mathcal{R}_h$ by the exact annulus $1 < |\zeta| \le 2$ can therefore change only points inside the $2Lh$ disks about 1 and 2. Their total transmission multiplicity is $O(M)$ by (4.7). This includes values lying exactly on a counting circle. The scalar endpoint contribution is also $O(M)$ by (2.5). Thus (5.1) becomes

$$N(\sqrt{2}/h) - N(1/h) = (2^{d/2} - 1)(\tau_1 + \tau_n)h^{-d} + O(h^{1-d}). \tag{5.3}$$

Here $\tau_1 + \tau_n = \omega_d(2\pi)^{-d}|\Omega|(1 + n^{d/2})$ is the combined scalar Weyl coefficient. Every sufficiently small real h is allowed: the auxiliary vertical sides were chosen for each h, and the local count removes their effect at the exact radii.

Proof of theorem 1.1. Put $c = \tau_1 + \tau_n$ and $h = \sqrt{2}/r$ in (5.3). For every sufficiently large real r,

$$N(r) - N(r/\sqrt{2}) = (1 - 2^{-d/2})cr^d + O(r^{d-1}).$$

Sum over $r, r/\sqrt{2}, r/2, \dots$ until the last radius belongs to a fixed bounded interval. The leading terms telescope, and the error sum is bounded by

$$Cr^{d-1} \sum_{j \geq 0} 2^{-j(d-1)/2} = O(r^{d-1}),$$

since $d \geq 2$. The remaining bounded-radius count is finite by lemma 2.1. Its contribution, and its leading volume term, are $O(1)$. Substitution of the two constants in (2.5) proves the stated formula. □

A Common Dirichlet values in a ball

The original algebraic multiplicity matters even in a simple radial example. Consider the unit ball in $\mathbb{R}^3$ with $n = 4$, restricted to radial fields. The map $(u, v) \mapsto (f, g) = (ru, rv)$ is unitary up to a fixed scalar, and the radial operator is

$$\mathcal{A}(f, g) = (-\tfrac{1}{4} f'', -g''), \qquad f(0) = g(0) = 0, \quad f(1) = g(1), \quad f'(1) = g'(1),$$

on $H^2(0, 1)^2$. The zero condition at the origin excludes the $1/r$ term, whose Laplacian has a delta mass and is not in the original graph domain. The radial subspace is a reducing block, since rotational averaging commutes with the operator and traces. This finite-interval realization has compact resolvent.

At energy $\lambda = k^2 \neq 0$, its solutions have the form $f = a \sin(2kr)$ and $g = b \sin(kr)$. The boundary matrix is

$$\begin{pmatrix} \sin 2k & -\sin k \\ 2k \cos 2k & -k \cos k \end{pmatrix}, \qquad \det = -2k \sin^3 k. \tag{A.1}$$

Compare the corresponding model determinant in [5, Section 2]. Thus the nonzero energies are $\lambda_m = (m\pi)^2$, $m \geq 1$, and each eigenspace has dimension one. We check its Riesz multiplicity directly rather than identify it merely with a determinant root order.

Near $\lambda_0 = k_0^2$, $k_0 = m\pi > 0$, choose the analytic root $k(\lambda)$. The analytic pair

$$W(\lambda, r) = \left(\sin(2k(\lambda)r), \frac{2 \cos(2k(\lambda))}{\cos k(\lambda)} \sin(k(\lambda)r) \right)$$

solves the differential equations and all but the value-matching condition. Its value mismatch is $2 \sin^3 k(\lambda)/\cos k(\lambda)$, of exact order three. The family is an identity for the differential expression, not an assertion that $W(\lambda)$ lies in the transmission domain for all λ. Its Taylor coefficients W_0, W_1, W_2 do lie in that domain and satisfy

$$(\mathcal{A} - \lambda_0) W_0 = 0, \quad (\mathcal{A} - \lambda_0) W_1 = W_0, \quad (\mathcal{A} - \lambda_0) W_2 = W_1.$$

There is no chain of length $\ell \geq 4$. Its formal Taylor polynomial would solve the two initial-value ODEs modulo $(\lambda - \lambda_0)^\ell$, hence have the form $a(\lambda) \sin(2k(\lambda)r), b(\lambda) \sin(k(\lambda)r)$ modulo that order. The leading eigenvector has $a(\lambda_0) \neq 0$. Derivative matching forces $b = 2a \cos 2k/\cos k$; value matching would then require $2a \sin^3 k/\cos k = O((\lambda - \lambda_0)^\ell)$, a contradiction. The one-dimensional eigenspace therefore belongs to one Jordan block of size three. Its original Riesz multiplicity is three.

For constant boundary values, the outward DN difference is

$$2k \cot 2k - k \cot k = -k \tan k.$$

At $k = m\pi$ its order is one, while $m_D = 2$ and $m_T = 3$; thus $1 = 3 - 2$ in (3.15). At a nonzero half-integer multiple of π it has a simple pole and $-1 = 0 - 1$. The exact radial count is

$$N_{\mathrm{rad}}(r) = 3\lfloor r/\pi \rfloor,$$

and the sum of the radial Dirichlet counts is $\lfloor r/\pi \rfloor + \lfloor 2r/\pi \rfloor$. Their difference is always zero or minus one, including thresholds. This calculation concerns only the radial block; it is not the full three-dimensional volume count and is not used in the proof of theorem 1.1.

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